

## Communication

|                      |                                                              |
|----------------------|--------------------------------------------------------------|
| <b>Date</b>          | 30 June 2026                                                 |
| <b>Team ID</b>       |                                                              |
| <b>Project Name</b>  | OptiCrop : Smart Agricultural Production Optimization Engine |
| <b>Maximum Marks</b> | 1 Mark                                                       |

### Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

| S.No | Communication Type    | Frequency | Channel / Tool                                | Participants     | Purpose                                                |
|------|-----------------------|-----------|-----------------------------------------------|------------------|--------------------------------------------------------|
| 1    | Self Project Planning | Daily     | Personal notes, GitHub                        | Bille Mrudula    | Plan daily development tasks and milestones            |
| 2    | Progress tracking     | Weekly    | Git & GitHub                                  | Bille Mrudula    | Track project progress and maintain version history    |
| 3    | Issue Resolution      | As Needed | ChatGPT, Python Documentation, Stack Overflow | Bille Mrudula    | Resolve coding, debugging, and machine learning issues |
| 4    | Stakeholder Review    | Bi-Weekly | Review Sessions                               | Student & Mentor | Receive feedback on project progress and improvements  |
| 5    | Final Demo Rehearsal  | Once      | PowerPoint & Local Flask Application          | Bille Mrudula    | Demonstrate the complete working application           |

### Communication Challenges & Resolutions:

| S.No | Challenge Faced                                                  | Resolution / Action Taken                                                                                                                                    |
|------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | Difficulty in selecting the most accurate machine learning model | Compared multiple algorithms (Logistic Regression, KNN, Decision Tree, and Random Forest) and selected Random Forest based on the highest accuracy (99.32%). |
| 2    | Flask backend integration with the trained model                 | Used Pickle to save and load the trained model and verified prediction functionality through local testing.                                                  |
| 3    | Input validation and user experience                             | Implemented both client-side (JavaScript) and server-side (Flask) validation to ensure reliable and user-friendly predictions.                               |